

SASHA LAMOT

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EDUCATION

University of California, Berkeley - B.A. in Computer Science

Graduation May 2025

Key coursework: Efficient Algorithms, Intro to AI and ML, Neural Networks, Database Systems, Computer Security, Discrete Mathematics and Probability Theory, Computer Architecture, Data Structures and Algorithms, Data Science and Statistics, Designing Information Devices and Systems, Data and Decisions in Business, Math - Calculus & Linear Algebra

International School of Brussels - International Baccalaureate (IB) Bilingual Diploma (44/45 Points)

Belgium June 2021

PROFESSIONAL EXPERIENCE

Apptweak (ASO Tool of the Year 2021) - App Store Optimization (ASO)

Jun - Aug 2024 | Brussels, Belgium

Software Engineer Intern

- Significantly **reduced loading time by 90%** on Apptweak's flagship market intelligence tracker, greatly **improving user experience**
- Implemented and deployed a **full database migration** into ClickHouse, with **fine-tuned structure** and enhanced data **insertion and retrieval mechanisms**, handling **~100 million rows** of data daily in a fraction of the time improving performance and support for rapid data growth

UC Berkeley Engineering Dept - Jadoo Technologies

Fall 2023 - Current | Berkeley, California

Software Engineer & Tech Lead in Dr. Waqas Khalid Lab

- Leading a team of 8** to develop software enabling **remote e-beam lithography** experiments for nanotech research
- Built **full-stack system** with a **frontend** and **backend** to simulate and run lithography processes, improving experiment control
- Integrated **web interfaces**, **Arduinos**, and **Picomotors**, designing an **algorithm** that exponentially increased lithography efficiency using multiple electron beams

IMEC (world-leading R&D and innovation hub in nanoelectronics and digital technologies)

Jun - Aug 2023 | Ghent, Belgium

Software Engineer Intern

- Worked collaboratively with researchers to **optimize data processing pipeline** for artificial nanopore experiments
- Built a pipeline to streamline flow from raw data to analysis, making it easier for scientists to derive insights from experiments
- Created and implemented **noise detection and mitigation algorithms** such as **CUSUM**, **mean drift**, **KDE plots**, to be used on **~10 million data points** per experiment, enhancing data accuracy and usability

Plan-it App

2018-2020 | Brussels, Belgium

Co-Founder & Chief Technology Officer

- Co-founded a personalized **ed-tech** app to **optimize student work** & study time for the International Baccalaureate
- Designed and developed **front-end** (HTML/CSS) and **back-end** (PHP, MySQL) systems, including secure login and data management
- Created the business model for scalability, recruited a team of 4, and liaised with IBO representatives.

PROJECTS

Forest Health Tracker

2024 | Berkeley, California

- Building software to **rate and track forest health** using **deep-learning for tree detection** and in-situ sensors to track forest floor conditions; can be used in wildlife restoration settings to automate tracking of forest health and regrowth progress

Machine Learning & Reinforcement Learning - Neural Networks

2024 | Berkeley, California

- Designed and coded NN with **PyTorch** for: number recognition, language translation, function approx. and game simulation
- Used methods such as **Deep Q-learning**, **Perceptrons**, **Value Iterations**, **Epsilon Greedy**

Gitlet

2022 Berkeley, California

- Built terminal version control system similar to **Git** with features such as **add**, **commit**, **merge**, **status**, **branch**, in Java

Attax

2022 Berkeley, California

- Strategy board game where player can play against **AI** created by using **Alpha-beta pruning** on a **game tree**

Remote Data Collection Module

2020-2021

Goa, India

- Designed and built a remote soil data collection system (air/soil temperature, humidity, sunlight) using **Arduinos**.
- Developed a **radio communication** system and **data streaming algorithm** for MySQL database storage.
- Created a **Java-based GUI** to graph, display, and manage data, with save/import functionalities and database integration.

SKILLS

Coding Languages: Python, Java, Ruby, C/C++, SQL, x86, HTML/CSS, Scheme, RISC-V, Regex

Libraries: PyTorch, Pandas, Numpy, Matplotlib, Flask

Tools: Kubernetes, Docker, API Development, AWS, Databases (PostgreSQL, Clickhouse, MySQL, Redshift), Git, Linux

General: Leadership, Public Speaking, Photography, Photoshop, Music, Eating Yummy Food

Languages: English, French, Dutch (Intermediate), Hindi (Beginner)